Combustion and Flame 158 (2011) 1469–1479

Contents lists available at ScienceDirect

Combustion and Flame

journal homepage: www.elsevier.com/locate/combustflame

Reduction of a detailed kinetic model for the ignition of methane/propane mixtures at gas turbine conditions using simulation error minimization methods

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ARTICLE INFO

Article history:
Received 24 August 2010
Received in revised form 6 December 2010
Accepted 8 December 2010
Available online 12 January 2011

Keywords:
Mechanism reduction
Skeletal mechanism
Natural gas
Simulation error minimization
Principal component analysis
Robustness analysis

ABSTRACT

Natural gas is the primary fuel for industrial gas turbines, which provide about one quarter of the world's primary energy supply. Beside methane it also contains larger hydrocarbons in small, varying ratios. This variation is expected to rise due to the increasing usage of non-traditional gas sources. Fuel composition has a large impact on auto-ignition delay time, which is a fundamental parameter for the optimal design and operation of gas turbines. For the oxidation of such mixtures, Curran, Petersen and co-workers recently developed a detailed reaction mechanism (NUIG NGM), which reproduces the ignition delays over a wide range of conditions. However, due to its large size: 229 species and 1359 reactions, it cannot be used in computational fluid dynamics simulations, which is an important fundamental tool in the development of gas turbines. A mechanism reduction case study of the NUIG NGM is presented using the recently developed simulation error minimization methods (SEM). A new version of the SEM program package is also proposed, which allows the reduction of mechanisms for a wider range of combustion phenomena. Combinational strategies have been introduced in the SEM connectivity method to enhance the reduction procedure and a hierarchical reduction procedure is proposed for multi-scenario problems. Ignition of lean and stoichiometric mixtures containing 90% methane and 10% propane as fuel were investigated for 22 conditions relevant to gas turbines, covering temperature and pressure ranges of 877–1465 K and 7–40 atm, respectively. The smallest reduced mechanism developed contains 50 species and 186 reactions. It can reproduce ignition delays with 3.1% maximum error and reproduces pressure rise precisely (error$\sim 10^{-3}$%). The mechanism can be simulated 62 times faster than the full mechanism. Robustness analysis showed that it is reliably applicable over a much wider range of conditions compared to that for which it was developed.

1. Introduction

1.1. Natural gas turbines

Natural gas is the primary fuel for industrial gas turbines [1], which currently provides about one quarter of the world's primary energy supply, and is expected to remain a significant source of energy until 2030. Natural gas is energy efficient and it has been used for several decades, thus it requires no new infrastructure to be built, unlike in the case of new sources of energy. Although natural gas is mostly methane (CH_4) it also contains larger hydrocarbons (ethane (C_2H_6), propane (C_3H_8), butane (C_4H_{10}), etc.) in varying ratios depending on its origin, extraction, and transport processes [2]. The variation in fuel composition is expected to increase due to the growing usage of non-traditional gas sources. This variation can impact the performance of dry, low-emission industrial gas turbine engines through variation in the auto-ignition delay time, flame dynamics, and flame speed [3–5].

Abbreviations: CM, connectivity method; CSP, computational singular perturbation; DAC, dynamic adaptive chemistry; DRG, directed relation graph; DRGASA, DRG-aided sensitivity analysis; DRGEP, DRG with error propagation; DRGEPASA, DRGEP-aided sensitivity analysis; DRGEPSA, DRGEP with sensitivity analysis; fPCA, functional principal component analysis; GA, genetic algorithm; HDMR, high-dimensional model reconstruction; ILDM, intrinsic low-dimensional manifold; ILP, integer linear programming; ISAT, insitu adaptive tabulation; KINALC, program package for the investigation of reaction mechanisms; NUIG NGM, natural gas mechanism developed at National University of Ireland, Galway; PCA, principal component analysis; PCAF, principal component analysis of the rate-sensitivity matrix; PRISM, piecewise implementation of solution mapping; QSSA, quasi-steady-state approximation; SEM, simulation error minimization; SEM-CM, connectivity method with simulation error minimization; SEM-PCAF, PCAF with simulation error minimization; TIBOX, integrator code of the SEM program package; TRANS, program for the conversion of CHEMKIN and TIBOX formats into each other.

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doi:10.1016/j.combustflame.2010.12.011

In the optimal design and operation of gas turbines, the ignition delay time (τ) of the fuel mixture is a fundamental parameter [6]. Reliable calculation of the ignition delay time, especially for the low-temperature regime, requires the application of very large detailed reaction mechanisms [7]. Curran, Petersen and co-workers have recently developed a reaction mechanism (NUIG Natural Gas Mechanism, NUIG NGM), which calculates ignition delays in good agreement with those measured for such natural gas mixtures over wide ranges of pressures (1–50 atm) and temperatures (740–1550 K) [8–15]. The sub-mechanism of C2–C3 hydrocarbon chemistry is the same as the recent natural gas [8–14], *n*-butane [16] and iso-butane [17] mechanisms and their blends [18] which have been published by the Combustion Chemistry Centre at NUI Galway.

In reacting flow simulations differential equations describing coupled transport and chemistry have to be solved, which makes them computationally very expensive. There are several computational fluid dynamics (CFD) codes [19], but most of them can handle only simple chemistry, which can have serious limitations [20]. Gas turbine combustor designers use special CFD software such as Fluent [21] or CHEMKIN-CFD [22], which allow the application of a detailed chemical kinetic mechanism. However, simulation time can easily become unacceptably long and there can be convergence problems with larger mechanisms, therefore a limit of 50 species was introduced in the Fluent code. There is no species limit in CHEMKIN-CFD, but it is not practical to include reaction mechanisms that contain more than 50 species. The size of the NUIG NGM is far larger (229 species, 1359 reactions) than can be used directly in CFD simulations.

1.2. Mechanism reduction

Almost all published detailed reaction mechanisms contain redundant species and reactions, because the developers want to make their detailed mechanisms as robust as possible, thus they include all possible analogous reactions which were previously found important in other systems, but may not necessarily be so in the new mechanism [23]. If a detailed reaction mechanism is applied for a special task (e.g. reproduction of ignition delays or pressure profiles at a defined range of conditions), then usually a significant number of species and reactions can be considered as redundant, that is they can be removed without significantly modifying simulation results. Mechanism reduction generally means a simplified mathematical description of chemical kinetics. The principal aims of reduction are to speed-up simulations and allow more complex CFD simulations than are otherwise possible for practical applications. These usually have non-zero spatial dimension or contain many coupled zero-dimensional reactors.

Mechanism reduction methods can be heuristic, which require problem-specific human effort, thus they cannot be generalized and automated by means of computers and this makes them non-applicable to large reaction mechanisms. Currently, systematic methods, which are based on well-defined mathematical procedures, are being used almost exclusively. A good summary on the systematic techniques of mechanism reduction methods can be found in the series "Comprehensive Chemical Kinetics" [23] and there are also a few reviews on the topic [20,24].

Systematic methods developed for mechanism reduction can be sorted into three major categories according to the level of simplifications introduced: there are (i) species and reaction elimination methods (see later in detail), (ii) lumping and (iii) solution mapping.

Lumping methods group species and reactions which are similar in chemical or physical properties and replace them with lumped species and reactions e.g. exact non-linear lumping [25], constrained lumping [26] and unconstrained lumping [27], approximate non-linear lumping based on time-scale analysis [28], reaction lumping using quasi-steady-state approximation [29].

Solution mapping methods replace mass-action rate based kinetic differential equations by their efficiently stored solution, thus mechanistic details get lost completely. Major solution mapping methods include (i) orthonormal polynomials [30,31], (ii) in situ adaptive tabulation (ISAT) [32], (iii) piecewise implementation of solution mapping (PRISM) [33] and (iv) high-dimensional model reconstruction (HDMR) [34] techniques.

Time-scale analysis plays a central role in mechanism reduction, as it can be the basis for methods of any of the three levels of reduction, and it can generally be used to find the lower dimensional approximation of a problem (computational singular perturbation (CSP) [20], quasi-steady-state approximation (QSSA) [20,29,35] and more general methods are computational singular perturbation (CSP) [36–39], and intrinsic low-dimensional manifold (ILDM) methods and its variants [30,31,40]).

1.3. Species and reaction elimination methods

The species and reaction elimination methods (or skeletal reduction methods) are highly popular and preferred by chemists, as they provide a subset of the species and reactions of the full mechanism whose chemical interpretation is straightforward. Furthermore, the kinetic and thermodynamic parameters of a skeletal mechanism can be given and used in the same way as in the full mechanism, thus they are readily applicable to popular codes such as CHEMKIN [41].

There are several skeletal reduction methods in the literature and new ones come to light almost every year (e.g. [35,36,38,39,42–61]). Elimination of species is more critical as it gives a major speed-up of simulation due to reduction in the number of variables in the kinetic differential equation system. While the simulation time of a reduced mechanism scales only linearly with the number of reactions, it is proportional to the number of non-zero elements in the kinetic Jacobian, which usually scales as the square of the number of species. Nevertheless, a significant speed-up can be achieved due to reaction removal, while the simulation error of the reduced mechanism remains essentially the same.

Species elimination is usually based on species-species connectivity measures, which can be obtained from the analysis of the Jacobian (also called connectivity method (CM) [42], connectivity method with simulation error minimization (SEM-CM) [43,44]), from rate of production analysis measures (directed relation graph (DRG) [45,62], DRG with error propagation (DRGEP) [46], DRG-aided sensitivity analysis (DRGASA) [63], DRGEP with sensitivity analysis (DRGEPSA) [47] or DRGEP-aided sensitivity analysis (DRGEPASA) [44]). Other species elimination methods can also be based on time-scale analysis (e.g. computational singular perturbation (CSP) [36]), element or atomic-flux analysis [61,64], etc.

Reaction elimination methods remove redundant reactions of previously determined non-redundant species. Few reaction elimination methods can eliminate redundant species and reactions simultaneously [36]. The major methods for the elimination of reactions include rate sensitivity analysis [48–50,53], principal component analysis of rate-sensitivity matrix (PCAF) [50–53] and its descendants (principal component analysis with simulation error minimization (SEM-PCAF) [43], functional PCA method (fPCA) [54]), rate of heat and species production analysis [55], computational singular perturbation (CSP) [37–39], genetic algorithm (GA) [57,58], integer linear programming (ILP) [56,59].

Methods can be sorted according to how many simulations they require per scenario. On-the-fly methods require no simulation with the full mechanism (e.g. dynamic adaptive chemistry (DAC) method is based on an extended version of DRG [60], atomic-flux analysis based method [61]).

Most of the mechanism reduction methods require one simulation with the full mechanism and no or very few simulations with the candidate reduced mechanisms per scenario. Whereas numerical optimization methods require several simulations of candidate reduced mechanisms, in return they can produce the smallest skeletal mechanism for a given problem (e.g. GA [57,58], SEM-CM and SEM-PCAF [43], ILP [56,59], DRGASA [63], DRGEPSA [47], DRGEPASA [44]).

1.4. Simulation error minimization methods

Species and reaction elimination methods with simulation error minimization (SEM-methods) [43,65] are based on Jacobian connectivity measures [42] and principal component analysis of the rate-sensitivity matrix (PCAF) [53], respectively. These methods, the simulation error minimization connectivity method (SEM-CM) and principal component analysis of matrix *F* with simulation error minimization (SEM-PCAF), were reported to produce very small and reduced mechanisms fulfilling accuracy requirements precisely.

Zsély et al. applied SEM methods with the minimization of concentration errors of major species to develop a reduced mechanism reproducing ignition delays [44]. They defined ignition delay as the time corresponding to the maximum of the pressure time-derivative. A moderate correlation was observed between the error of ignition delays and the root mean square (RMS) error of the concentration profiles. The new version of SEM program package was made better suitable for ignitions problems [64]. In this paper we applied SEM methods to the NUIG NGM to develop skeletal mechanisms, which can also accurately reproduce ignition delays for methane/propane mixtures under gas turbine conditions.

1.5. Robustness analysis

The SEM algorithms seek the smallest reduced mechanism, whose simulation results agree with those of the original mechanism with a given accuracy on a set of selected scenarios. *A posteriori*, it is interesting to investigate whether the mechanism is reliable at other conditions, too. In the multidimensional space of initial conditions the valid ranges for a reduced mechanism are found to be non-convex and sometimes even disjoint [58,59,66,67]. Therefore even if a reduced mechanism is minimal in size among all possible reduced mechanisms for a hyperrectangular range of conditions, it probably has good predictive power at other conditions, too. Only a few of the mechanism reduction studies investigate the range of validity of their reduced mechanisms [58,59,67]. To determine these valid ranges robustness analysis is applied, which investigates the deviation of simulation results of the full and reduced mechanisms for a densely sampled set of initial conditions.

In Section 2, the improvements in the SEM methods and SEM program package are introduced. In Section 3, the details of a mechanism reduction case study on the NUIG natural gas mechanism are discussed. In Sections 4 and 5, the results of mechanism reduction and the results of the robustness analysis are presented, respectively.

2. Simulation error minimization methods

In this section, the essentials of the SEM methods are summarized to help in the understanding of new developments. For complete understanding of the SEM methods please check reference [43] and webpage [65]. The old and the new versions of the SEM program package with manual is made available on the World Wide Web [65].

2.1. Error definition

For the development of a reduced mechanism, which is applicable for more scenarios, a definition of one single global error is required as a function of scenario errors. In the SEM procedures, advancement and efficiency of the reduction process was monitored using global maximum (max) and root mean square (rms) errors of the concentrations of the important species [43]. While minimization of rms error is less prone to failure, it is based on a less reliable error measure, because a small rms error allows extremely large errors for a small number of cases. The integrator of the SEM package (TIBOX) allowed the selection of times for local error calculations to be distributed logarithmically in a scenario [65]. At given numbers of species and reactions, reduced mechanisms with smallest global errors, both max and rms, are sought in SEM methods.

2.2. SEM-CM method – species removal

The recently developed SEM-CM algorithm [43] is based on the connectivity method [42], but solves many of its associated problems. The SEM-CM method defines the terms "complementary set of species", "living species" and "consistent mechanism" to help its elucidation [43]. The SEM-CM method starts from a set of important species, which has to be defined by the user for each scenario. "Complementary set of species" is defined as those species not yet selected, whose addition to the group of currently selected species would make at least one additional reaction selected. A reaction becomes selected if all of its species are selected. A species is called a "living species" in a scenario if its initial concentration is non-zero or it has an inflow term (e.g. non-zero inlet concentration in a PSR) or if it is formed in a chemical reaction. The list of living species depends on the mechanism and also on the initial (or boundary) conditions. A mechanism is called "consistent" if each of its species is living at least in one of the scenarios.

Starting from the group of *n* important species, the first *m* complementary sets (*m* is the depth level of investigation) that are most connected to them are determined at each selected time. Direct kinetic connectivity of the ith species to the jth species is characterized by the corresponding log-normalized Jacobi matrix element $\bar{\mathbf{J}}_{ij}$, which is defined as $\bar{\mathbf{J}}_{ij} = c_i/f_j \cdot \partial f_j/\partial c_i$, where c_i and f_j denote the concentration and the rate of production of the species ith and jth respectively. The strength of the direct link of the complementary set k (containing n_k species) to the group of selected species is characterized by the following quantity:

$$C_k = \frac{1}{n_k} \sum_{j \in \text{complementary set k}} \; \sum_{i \in \text{selected group}} \bar{\mathbf{J}}_{ij}^2 \tag{1}$$

Most connected sets are added to the set of important species one by one and in each case a reduced mechanism is formed. These reduced mechanisms are tested for consistency, and made consistent if necessary by adding additional complementary sets [43]. The complex model is simulated with each of the candidate mechanisms and their sets of species with their errors are stored in a database.

A new cycle is initiated by searching the database for the two reduced mechanisms that have the smallest max and rms errors among the ones having *n* + 1 species in the database. If no mechanism having *n* + 1 species exists in the database then the one with species *n* + 2 is sought, and so on until one is found. The building step described in the previous paragraph is repeated with both of them, until max error of a reduced mechanism does not become smaller than the user-prescribed error threshold. This way, a series of consistent reduced mechanisms are produced, usually with continuously decreasing error. Repeating the whole procedure with gradually increasing (e.g. doubling, quadrupling) depth level of investigation (m) and utilizing the databases created by previous runs, and usually smaller and smaller reduced mechanisms are obtained which fulfil the accuracy criteria.

2.3. SEM-PCAF method – reaction removal

The SEM-CM procedure results in a series of consistent reduced mechanisms with a different number of necessary species. These mechanisms can be reduced further via the elimination of redundant reactions. The principal component analysis (PCA) of the normalized rate-sensitivity matrix $\overline{\mathbf{F}}$ (PCAF) [53] is an efficient way of identifying necessary reactions. The elements of the log-normalized rate-sensitivity matrix are defined as $\overline{\mathbf{F}}_{ij} = k_i/f_j \cdot \partial f_j/\partial k_i$, where k_i denotes the rate coefficient of the ith reaction. This method has been encoded as the PCAF option of KINALC [68]. In the PCAF method the indices of large eigenvector components of the large eigenvalues of the $\overline{\mathbf{F}}^T\overline{\mathbf{F}}$ matrix correspond to the reactions which are necessary to accurately reproduce the concentration profiles of species in the reduced mechanism.

In the program KINALC the user suggests one or several pairs of thresholds (lower bounds) for the eigenvalues and the eigenvector components, and based on these, the corresponding reactions are selected. In the principal analysis of the normalized rate-sensitivity matrix with the simulation error minimization method (SEM-PCAF) [43,65] the PCAF procedure [50–53] is adapted in such a way that many different thresholds are tested automatically and the corresponding reduced mechanisms are formed. Only consistent reduced mechanisms are investigated further and the one with the smallest simulation time which still fulfils the accuracy requirements is selected.

2.4. Extension of SEM program package for ignition problems

Ignition delays are usually determined in shock tube or rapid compression machine experiments, which can, most of the time, be simulated as an isochoric-adiabatic process or as an isochoric process with heat loss in the post-compression phase, respectively. The proper simulation of heat loss is required in order to validate the full mechanism against experimental measurement, whereas for mechanism reduction purposes, isochoric-adiabatic conditions are adequate to test the reduced mechanisms against the full mechanism. Integrator code TIBOX was extended to the simulations of adiabatic and isochoric conditions, which permit the study of a wider range of combustion phenomena. A new version of the code TIBOX predicts the same pressure and temperature profiles as program CHEMKIN-PRO [41]. In the new version of the SEM package, times for error calculation and reduction analysis can also be selected in a linear fashion, which is more appropriate for ignition problems, where significant changes usually occur within one order of magnitude of time.

At isochoric-adiabatic conditions, ignition delay was defined as the time belonging to the largest first derivative of the pressure–time profile. Some of the reduced mechanisms predicted the time of pressure rise close to that of the full mechanism, whereas the predicted pressure rise was negligible, thus the system was not considered to have ignited properly. Therefore, another requirement was defined: the largest pressure time-derivative should be at least one hundredth of the largest pressure time-derivative calculated by the full mechanism. A simple relative error of ignition delay for scenario i (denoted by $\delta_{\tau i}$) was used for monitoring the error of the reduced mechanisms. In the new version of the SEM package, several new global errors were introduced other than simple root mean square and simple maximum errors [65].

The two optimization threads of the SEM procedure minimize rms and max errors in parallel. The following global errors for ignition delay are available in the new version of the SEM package:

$$\begin{aligned} \delta_{\tau\ \mathrm{MAX}} &= \max_{\mathrm{i:scenarios}} \delta_{\tau\ \mathrm{i}} \\ \delta_{\tau\ \mathrm{RMS}} &= \operatorname*{rms}_{\mathrm{i:scenarios}} \delta_{\tau\ \mathrm{i}} \end{aligned} \qquad (2)$$

The direct introduction of the ignition delay time and the pressure profile in the error function used in the SEM methods overcomes the problem of the selection of important species related to the ignition phenomena addressed in several papers [44,47,60]. For our calculations we used rms and max global errors of the ignition delays in all scenarios.

2.5. Extension of the SEM-CM method by combinational complementary sets

A simple complementary set is defined for each unselected reaction, by taking its species and neglecting those which are already selected. If several scenarios are considered simultaneously, it is not always possible to decrease the global error of a reduced mechanism continuously by adding a simple complementary set to it, even if all possible simple complementary sets are tested. In this case, the addition of the union of different complementary sets that we call a complex complementary set, should be investigated to reduce the error of the reduced mechanism further. If we have m simple complementary sets, then the number of the possible two-combinations of them is $m\cdot(m-1)/2$. The number of different sets is usually less, but is in the same order of magnitude. Combining three, four, and even more simple complementary sets the number of combinations quickly becomes very large. Since the SEM-CM method requires the full simulation of each candidate mechanism formed by the addition of complementary sets investigating higher-combinations is not practical.

To confine the combinations only to a certain number of strongly connected complementary sets we use four integers ($c_{\min}$, $c_{\max}$, $d_{\max}$, $s_{\max}$) to define a simple strategy. Value $d_{\max}$ designates simple complementary sets from the first to the $d_{\max}$-th in the connectivity rankings of complementary sets determined at each investigation time. Note, it has the same meaning as the depth level in the original version of SEM-CM. These first $d_{\max}$ simple complementary sets are conflated to form complex complementary sets. Values $c_{\min}$ and $c_{\max}$ denote the minimum and the maximum number of simple complementary sets to be combined (this implies $c_{\min} \leqslant c_{\max} \leqslant d_{\max}$). The value of $s_{\max}$ can be used to reduce the number of large candidate mechanisms by limiting the number of added species. This is a very useful constraint, as large additions were found to be less effective in decreasing the error and they may require substantially more simulation time. In the new version of the program, several simple strategies can be applied simultaneously to form complex strategies.

Let us apply (1, 1, 6, 4) + (2, 2, 5, 4) + (3, 3, 5, 5) complex strategy on the rank list of complementary sets of an artificial example. Table 1 shows the rank of simple complementary sets, the simple and complex complementary sets obtained with simple strategies and with the complex strategy.

3. Mechanism reduction case study

3.1. Investigated combustion system

Curran, Petersen and co-workers recently developed a detailed reaction mechanism, the National University of Ireland Natural

Table 1
Example for simple and complex combinatorial strategies. Different capital letters denote different species. "Words" formed by them with alphabetical ordered letters denote simple complementary sets. In columns of simple and complex strategies, list of complementary sets are shown allowed by the strategy. Complementary sets generated by more than one simple strategy are written in bold. Complementary sets that have more species than allowed by the strategy are struck out. These are not listed in the column of complex strategy.

Rank of simple complementary sets	Simple strategies			Complex strategy (1, 1, 6, 4) + (2, 2, 5, 4) + (3, 3, 5, 5)
	(1, 1, 6, 4)	(2, 2, 5, 4)	(3, 3, 5, 5)	
1. A	A, BC, CDE,	ABC, ACDE,	ABCDE, ABCF,	A, BC, CDE, F, **FG**, EFG, ABC,
2. BC	F, **FG**, EFG	AF, **AFG**,	ABCFG, ACDEF,	ACDE, AF, **AFG**, BCDE, BCF,
3. CDE		BCDE, BCF,	~~ACDEFG~~, **AFG**,	**BCFG**, CDEF, ABCDE, ABCF,
4. F		**BCFG**, CDEF,	BCDEF, ~~BCDEFG~~,	ABCFG, ACDEF, BCDEF, CDEFG
5. FG		~~CDEFG~~, **FG**	**BCFG**, CDEFG	
6. EFG				

Gas Mechanism (NUIG NGM) to describe the oxidation of such mixtures [8–14]. The latest version of NUIG NGM (version III or mechanism C4_49) [15], which contains 1295 reversible and 64 irreversible reactions of 227 reactive species up to 4 carbon atoms, and 2 non-reactive species (N_2, Ar), was reduced by the SEM methods. The SEM methods require the reactions of the mechanism to be in irreversible form; hence all reactions in the original mechanism were converted to irreversible pairs using the program MECHMOD [69]. Note that the program MECHMOD does not handle those reversible reactions properly where Arrhenius parameters for the backward reaction are given by the REV keyword. To avoid this problem one is required to define two irreversible reactions instead of using the REV keyword. The new version of code Trans, which is part of the SEM package, can also convert reversible reactions to irreversible pairs [65].

The scenarios of simulations, where the reduced mechanism was expected to reproduce the results of the original mechanism, are based on 22 selected experiments published in the paper of Healy et al. [11]. In these experiments, ignition delays were measured for lean ($\phi = 0.5$) and stoichiometric ($\phi = 1$), 90% methane and 10% propane mixtures diluted in Ar and N_2 in ratios of about 1:2.6 and 1:3.1, using a rapid compression machine at lower temperatures and a shock tube at higher temperatures. These experiments cover gas turbine relevant conditions, with temperatures in the range of 877–1465 K and over a pressure range of 7–40 atm. The 22 scenarios were grouped into 6 regimes based on experimental technique, and pressure and composition. Details of scenarios and regimes are shown in Table 2.

3.2. Initial selection of species in the SEM-CM procedure

The aim of the current study is to develop a reduced mechanism that can reproduce ignition delay and pressure rise with good accuracy over a wide range of conditions. The SEM algorithm requires the selection of an initial set of species in order to initiate the reduction procedure. Thereafter, additional species are selected based on the strength of the kinetic connectivity of other species to them. The connectivity measure defined by an element of the log-normalized Jacobi matrix ($\bar{\mathbf{J}}_{ij}$) is applicable only for species to species connection, thus the direct use of pressure rise and the ignition delay time is not possible. The concentrations of major fuel and oxidant species change dramatically during ignition, so they can be used to characterize the ignition event. Their concentration profiles are only indirectly related to the ignition delay time and

Table 2
Regimes and scenarios used for the mechanism reduction. Scenarios are based on ignition delay measurements in rapid compression machines (RCM) and shock tubes (ST) carried out by Healy et al. [11].

Regime	Scenario	Experiment	Initial conditions		Fuel mixture		ϕ (O_2)	Diluent mol%	
			p/atm	T/K	CH_4: C_3H_8			Ar	N_2
1	1	RCM	9.79	1033	9	1	0.5	75.55	0
	2		10.44	1099					
2	3		19.77	955					
	4		20.11	1037					
	5		20.44	904				37.78	37.78
	6		21.86	916				75.55	0
3	7	ST	29.40	1369					
	8		30.60	1252					
	9		30.90	1294					
4	10	RCM	39.10	909				37.78	37.78
	11		39.44	877					
5	12	ST	6.84	1318			1	0	72.39
	13		8.54	1465					
	14		8.94	1281					
6	15		15.06	1234					
	16		16.01	1142					
	17		17.46	1253					
	18		18.00	1271					
	19		21.76	1219					
	20		22.83	1193					
	21		23.34	1170					
	22		24.32	1117					

Table 3
The result of mechanism reduction with SEM-CM procedure for each scenario is shown. Species set of scenario-level reduced mechanisms are shown with their maximum simulation error in ignition delay time. Labels of species are shown which are common in all scenarios specific to each regime and specific to each scenario, respectively.

Regime	Scenario	Reactive species in the scenario-level reduced mechanisms				$\delta_{\tau,i}$%
		#	Common (+Ar/N_2)	Regime specific	Scenario specific	
1	1	14	CH_4, C_3H_8, O_2	HO_2, CH_3O	H_2O_2, C_2H_4, nC_3H_7	1.0
	2	11	H_2O, H, O, OH, CH_2O, CH_3		–	3.0
2	3	15		H_2	CO, HCO, CH_3O, CH_3, C_2H_5	4.1
	4	15			C_2H_4, iC_3H_7, C_3H_6, C_3H_5-s, iC_4H_{10}	~0.1
	5	18			CO, HCO, CH_3OH, CH_3O, CH_3O_2H CH_3O_2, C_2H_5, nC_3H_7	3.0
	6	17			CO, HCO, CH_3O, CH_3O_2, C_2H_5, C_2H_4, C_2H_5O	~0.1
3	7	12		–	nC_3H_7, C_3H_6, C_3H_6OH	5.0
	8	12			HO_2, H_2O_2, CH_3O	5.0
	9	12			H_2, CH_3O, CH_3O_2	0.8
4	10	14		H_2, CO, HCO, CH_3O, C_2H_5	–	~0.1
	11	14			–	4.0
5	12	14		–	HOCHO, $HOCH_2O$, CH_3O, C_2H_5, C_2H_4O1–2	0.1
	13	12			C_2H_6, C_2H_5, C_2H_4, C_2H_5OH (no CH_2O!)	2.1
	14	12			H_2, iC_3H_7, C_3H_6	3.3
6	15	11		–	HO_2, CH_3O	4.1
	16	12			CH_3O, C_2H_5, C_2H_4O1–2	4.0
	17	12			HO_2, H_2O_2, CH_3O	~0.1
	18	12			CH_3O, C_2H_5, C_2H_4O1–2	0.3
	19	12			CO, HCO, CH_3O	~0.1
	20	12			CH_3O, C_2H_5, C_2H_4O1-2	0.2
	21	13			CH_3OH, iC_3H_7, C_3H_6, C_3H_5-s	4.0
	22	12			H_2, iC_3H_7, C_3H_6	0.4

Table 4
The result of the reduction with SEM-CM procedure. Species sets of reduced mechanisms for regimes are shown with their simulation error in ignition delay time. Labels of species are shown which are common in all scenarios, specific to each regime and specific to each scenario, respectively.

Regime	Reactive species in the regime level reduced mechanisms			$\delta_{\tau max}$**%**
	#	Common (+Ar/N_2)	Regime specific	
1	18	CH_4, C_3H_8, O_2	CO, CH_3O, CH_3O_2H, CH_3O_2, CH_2(s), C_2H_5, C_2H_5OH, nC_3H_7, $nC_3H_7O_2$	0.2
2	18	H, O, OH,	CO_2, HCO, HO_2CHO, O_2CHO, HOCHO, OCHO, $HOCH_2O$, CH_3O, C_2H_5	3.9
3	18	H_2O	C_2H_5, C_2H_4, CH_3CHO, nC_3H_7, C_3H_6, C_3H_5-a, C_3H_4-a, C_4H_8-1, C_4H_71-2	1.1
4	14	CH_3, CH_2O	CO, HCO, CH_3OH, CH_3O, CH_3CHO	2.0
5	21		H_2, CH_3OH, CH_3O, CH_2(s), C_2H_5, C_2H_4, C_2H_3, CH_3CO, CH_2CHO, CH_2CO, C_2H_3O1–2, iC_3H_7	2.7
6	27		H_2, HO_2, H_2O_2, CO, HCO, CH_3O, CH_3O_2H, CH_3O_2, C_2H_5, CH_3CHO, C_2H_5O, iC_3H_7, nC_3H_7, $nC_3H_7O_2H$, $iC_3H_7O_2H$, $nC_3H_7O_2$, $iC_3H_7O_2$, nC_3H_7O	4.1

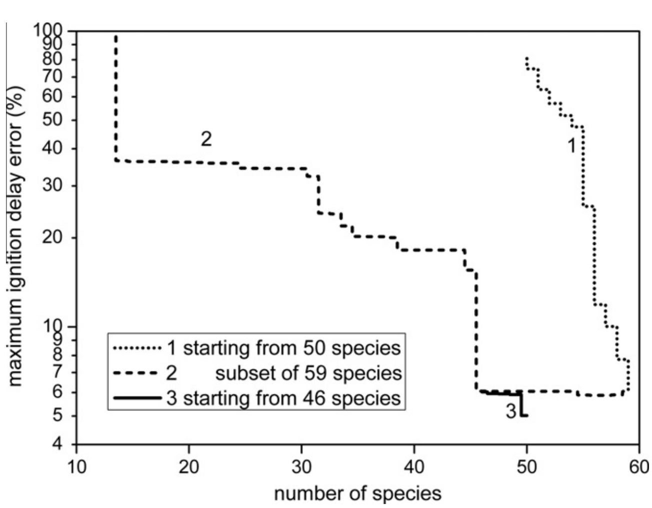

Fig. 1. Evolution of maximum ignition delay error during three SEM-CM procedures. The first procedure builds the 50-species set mechanism further to a 59-species set having 6% error. The second step searches for optimal subset of the 59-species set and a 46-species set is found with 6% error. In the third procedure starting from the 46-species set the ignition delay error is reduced further down to 5% by adding 4 species. To aid the eyes lines are shown which are built-up from connected points at integer species numbers.

the pressure rise, nevertheless they are good candidates to initiate the reduction procedure. In this work, the major reactant species (CH_4, C_3H_8 and O_2) were selected initially. Note, depending on the problem, also major products and even major intermediates can be selected initially.

3.3. Error types used for reduction and aimed error thresholds

The possible maximal error in simulated ignition delay time of a reduced mechanism is limited by the length of simulation in a scenario. To obtain uniformly defined, thus comparable errors for different scenarios, the simulation time interval of each scenario was set to be twice its ignition delay time (τ). The linear distribution of l points on a 2τ period guarantees $2\tau/l$ time resolution, which limits the maximum error in ignition delay determination to $100/l$%. Furthermore, interpolation is used to estimate the ignition delay time, thus the error is usually much less. To develop a reduced mechanism with 5% or less error in ignition delay $l = 100$ points were used in the error calculations and the connectivity was investigated in every 5th point.

The ignition delay time can vary within several orders of magnitude from a few μs to a couple of hundred ms depending on the circumstances. The NUIG NGM is capable of predicting experimental ignition delays of such mixtures with accuracy at

Table 5
Characteristics of the mechanisms. Reactions are classified as reversible (rever.) or irreversible (irrever.). Increase in simulation speed of reduced mechanisms is compared to that of the full mechanism. Maximum (MAX) and root mean square (RMS) of relative errors of ignition delays and pressure rises are shown. Errors were calculated precisely by using 1000 points in time range of $\tau/2$ and 2τ in logarithmic manner. Note, speed-up was determined from simulations without the overhead of detailed saving of simulation results.

Mechanism	Species (Ar/N_2)	Number of reactions			Speed-up of simulations ($\times$times)	Relative error in ignition delays (%)		Relative error in pressure rises (%)	
		Rever.	Irrever.	Total		Max	Rms	Max	Rms
NUIG NGM	229	1295	64	1359	–	–	–	–	–
RM1	50	221	30	251	45	4.9	2.9	0.0011	0.0007
RM2	50	97	89	186	62	3.1	2.0	0.0033	0.0016

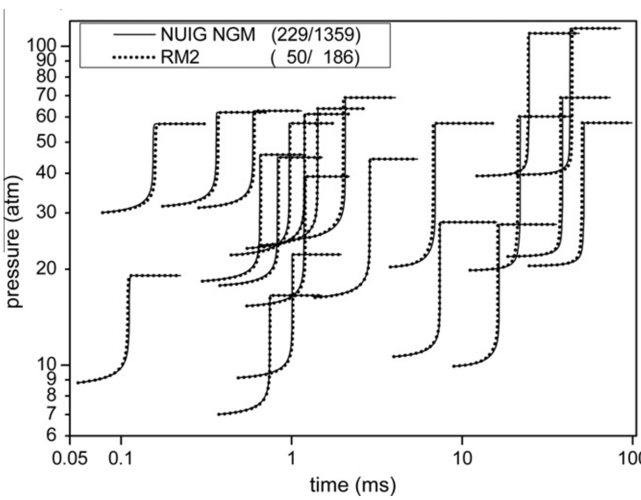

Fig. 2. The performance of the reduced mechanism obtained after species and reaction removal with SEM procedures. The reduced mechanism is capable of reproducing the ignition delay times accurately, within 3.1% error, and pressure rises precisely, within 10^{-5}% error for all the 22 scenarios. Pressure profiles are shown only for times between $\tau_i/2$ and $2\,\tau_i$ for better transparency.

intermediate and high temperatures. To provide a reduced mechanism with similar predictive power, a maximum error of 5% in the ignition delay times was set as requirement during the reduction. The accurate prediction of the pressure rise can also be an important feature of the reduced mechanism depending on the application (e.g. in HCCI simulations). Surprisingly, the mechanism developed for ignition delays also proved to be accurate in predicting pressure rise. The reasons for this may be the wide spectrum of circumstances and all species which are important for the full heat release in the scenarios must play an important role in the heat release at the beginning of ignition for at least one of the scenarios.

3.4. Hierarchical species elimination procedure for multi-scenario problems

The SEM-CM algorithm tries to fulfil a very strong requirement: it generates the reduced mechanisms by forcing a continuous decrease of simulation error. When one tries to minimize the maximum error of many scenarios simultaneously with the SEM-CM procedure, a large number of combinations of complementary sets are required to reduce the error to small values in a steep manner. To ease this task a hierarchical procedure is proposed.

Firstly, starting from the major species, very accurate reduced mechanisms, with less than 5% error in ignition delay, were developed for each scenario by applying complex strategy (1, 1, all, 6) + (2, 2, 50, 6) + (3, 3, 20, 6). The notation "d_{max} = all" means that all possible simple complementary sets were tested. This complex strategy made it unnecessary to run the SEM-CM procedure with increasing depth levels, as originally suggested in the first version of SEM-CM. For each regime the intersection of the species lists of reduced mechanisms obtained for the scenarios of the regime. A second SEM-CM procedure was carried out with the same complex strategy for each regime starting from the corresponding intersection set aiming at 5% error in all scenarios. Thereafter the species lists obtained for the regimes were unified and starting from this set a third SEM-CM procedure with complex strategy (1, 1, all, 10) + (2, 10, 10, 20) was performed aiming at 5% error. Note, in this case starting from the intersection of the species set would have been highly expensive computationally.

4. Reduced mechanisms

Species of the reduced mechanisms obtained for the scenarios by the first SEM-CM run are shown in Table 3. After careful examination of them, a common set of species can be extracted which appear in all reduced mechanisms: the diluent (Ar and/or N_2) and the fuel-oxidant mixture (CH_4, C_3H_8, O_2) and the species H, O, OH, H_2O, CH_3. Formaldehyde (CH_2O) also appears in all scenarios except one, thus it can be considered as a globally common species. Note, for some of the scenarios no propane reactions are required to predict the ignition delay under those conditions. Species of the reduced mechanisms obtained for the regimes by the second SEM-CM run are presented in Table 4.

The lists of species from all regimes were unified and the corresponding 50-species mechanism was assembled and simulated for all scenarios, giving 81% maximum error in ignition delays. The third reduction stage was started from this 50-species set with a complex strategy (1, 1, all, 10) + (2, 10, 10, 20) adding species from the full species set. The maximum error in ignition delays could be reduced very rapidly and after the inclusion of 9 species ($a$$C_3H_5OOH$, C_2H_6, $C_3H_5$2-1_3OOH, C_3H_5O, C_3H_6O1-3, C_3H_6OOH1-3, C_3H_6OOH1-3O_2, C_3ket13) it reached 6.1% error at 59 species, where the reduction procedure was stopped.

The SEM-CM code also allows selection of species from a subset of species instead of the full set, while minimizing the simulation error obtained by taking the full mechanism as reference. The best

Table 6
Test matrix used to determine the robustness of the reduced mechanisms. Simulations are done for all the combinations of scenario parameters listed in the table. Number of different values of each scenario parameter is shown in the brackets. Altogether there are 13 temperatures, 13 pressures, 3 fuel compositions, 3 equivalence ratios, 2 dilutions and 3 diluent compositions, which generates 9126 scenarios.

T (K) (13)	p (atm) (13)	CH_4:C_3H_8 ratio (3)	φ (O_2) (3)	Diluent Ar:N_2 ratio (3)	Dilution (diluent mol%) (2)
800, 900,..., 2000	1, 2.5, 5, 7.5, 10,15,..., 50	10:0, 9:1, 8:2	0.5, 1.0, 2.0	1:0, 1:1, 0:1	75%, 95%

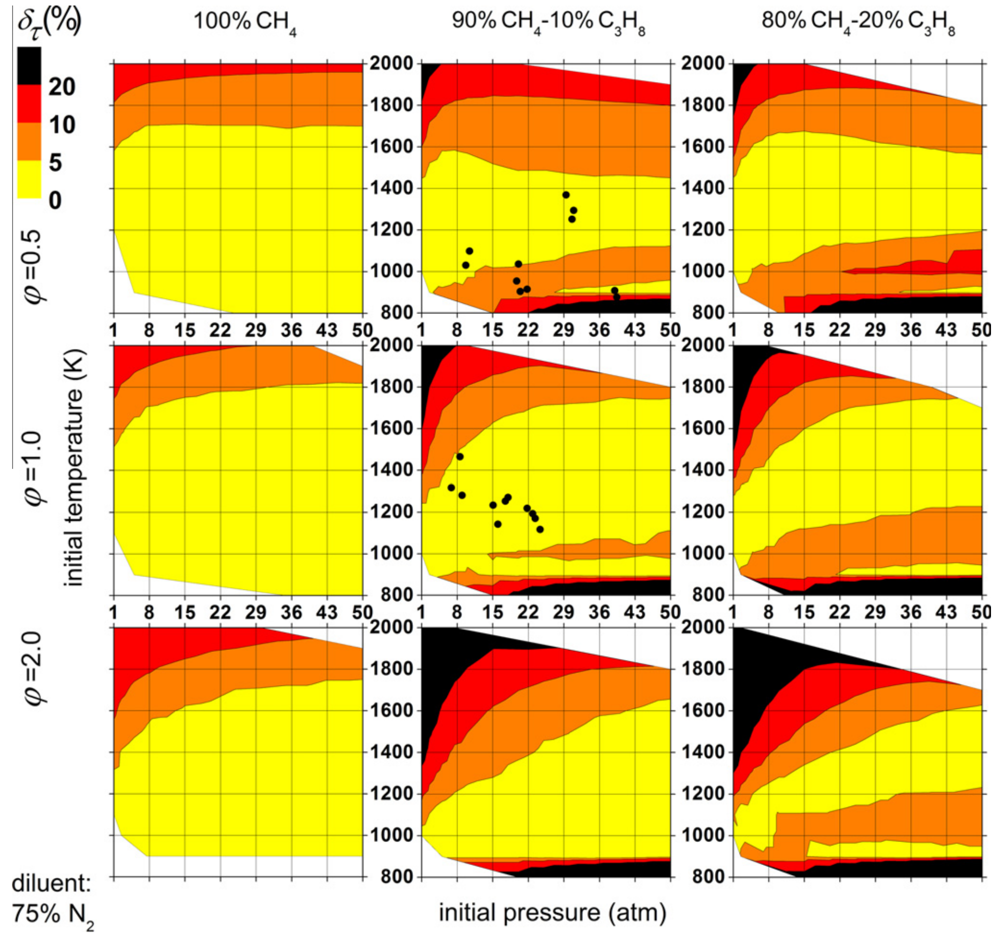

Fig. 3. Relative error of ignition delay simulated with reduced mechanism RM1 is shown as a function of initial pressure and temperature for various fuel compositions and O_2 equivalence ratios with 75% N_2 diluent. White area belongs to scenarios where ignition delays calculated by the full mechanism were outside of 2 μs–500 ms range. The dots denote the scenarios of mechanism reduction, whose initial mixture compositions are slightly different from those used in robustness analysis, therefore they might appear as having an error larger than previously stated.

subset of the 59-species mechanism was sought by starting a reduction procedure with a complex strategy (1, 1, all, 10) from the 11-species common set of regimes. A 46-species subset was identified (by removal of species aC_3H_5OOH, C_2H_5O, C_2H_5OH, C_3H_4-a, $C_3H_5$2-1_3OOH, $C_4H_7$1-2, CH_2(s), CH_3OCH_3, CH_3OH, HO-CH_2O, HOCHO, nC_3H_7O, $nC_3H_7O_2H$), which was also capable of predicting ignition delays within 6.1% error.

This 46-species mechanism was built further with complex strategy (1, 1, all, 10) + (2, 10, 10, 20) and its error could be reduced steeply to 5% by adding only 4 species (CH_3OH, C_2H_4O1-2, $C_3H_5$2-1_3OOH, aC_3H_5OOH), where it started stagnating again. This mechanism had 50 species, 221 reversible and 30 irreversible reactions (reduced mechanism RM1). The evolution of the maximum error in ignition delay as a function of the number of species in the reduced mechanisms during the last three SEM-CM procedures are shown on Fig. 1.

The SEM-PCAF procedure was applied to remove redundant reactions and after several restarts it provided a mechanism with 97 reversible and 89 irreversible reactions (reduced mechanism RM2). This mechanism can be simulated, on average, 62 times faster and can reproduce ignition delays within 3.2% error, and predict pressure rise very precisely for all of the 22 scenarios. The performance and properties of the mechanisms are shown in Table 5. The excellent agreement between the pressure profiles calculated by the detailed mechanism and the final reduced mechanism for the 22 scenarios are shown in Fig. 2. In Section 1.1, it was mentioned that the practical limit in mechanism size regarding the number of species in CFD simulations is 50 species, which was exactly matched by the accurate reduced mechanisms developed by the SEM procedures. Calculations took two weeks on one core of a modern 3 GHz PC.

5. Robustness analysis of the reduced mechanisms

A robustness analysis was carried out to determine at what other physical conditions and compositions the reduced mechanisms can reliably (e.g. with less than 20% error) estimate the results of the full mechanisms. Robustness of the reduced mechanisms was investigated over a range of conditions wider than those over which the NUIG NGM was validated. It is questionable to examine robustness of a reduced mechanism outside its proven range of validity, since its usage there is just as unjustified as that of the full mechanism. However, detailed mechanisms incorporate the best of our current knowledge, thus they are frequently used for simulations at conditions where validation has not been available or not yet possible. This practice justifies such a wide robustness study. Nearly 10,000 scenarios were defined by combining temperature, pressure, methane-propane ratio, equivalence ratio, diluent composition and dilution values shown in Table 6.

For both the full and reduced mechanisms the simulation time was set to 1s, and outputs were generated from 1 μs to 1 s at 1500

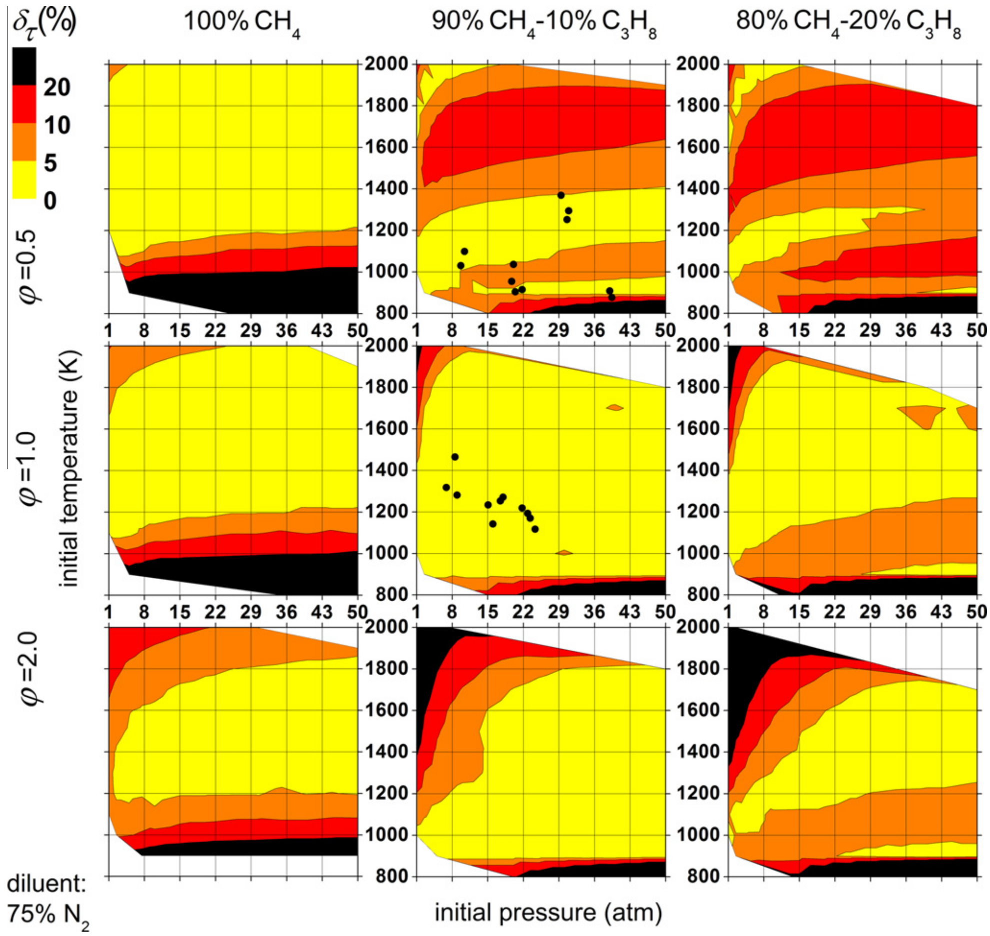

Fig. 4. Relative error of ignition delay simulated with reduced mechanism RM2 is shown as a function of initial pressure and temperature for various fuel compositions and O_2 equivalence ratios with 75% N_2 diluent. White area belongs to scenarios where ignition delays calculated by the full mechanism were outside of 2 μs–500 ms range. The dots denote the scenarios of mechanism reduction, whose initial mixture compositions are slightly different from those used in robustness analysis, therefore they might appear as having an error larger than previously stated.

points in logarithmic distribution to capture ignition delays precisely at every order of magnitude. To give correct errors for both under- and over-estimation by a factor of no more than 2 (up to 50% error), only results of those conditions at which the full mechanism predicted an ignition delay between 2 μs and 500 ms are plotted.

The detailed and reduced mechanisms were simulated for all the scenarios. Ignition delays and pressure rises were determined and their relative errors were calculated (all available in the supplementary material). Only results belonging to mixtures containing 75% N_2 dilute gas are presented and discussed in detail in this paper, as this condition is the one closest to practical applications.

Figs. 3 and 4 show the error in ignition delay simulated by reduced mechanisms RM1 and RM2, respectively. Conditions closest to the scenarios used for mechanism reduction are shown on the graphs as dots. Note that the diluent compositions and mixing ratios used in the scenarios for mechanism reduction and ones investigated in the robustness analysis are slightly different therefore the corresponding dots might appear outside of the 5% error range.

Mechanism RM1 can predict ignition delays calculated by the full mechanism within 10% error for methane-propane ratios of 10:0, 9:1, 8:2 and O_2 equivalence ratios of 0.5, 1.0 and 2.0 at the pressures and temperatures of the 22 scenarios. Ignition delay time of pure methane fuel is accurately (within 10% error) predicted except for very high temperatures (above 1900 K). The agreement for methane gets slightly worse for equivalence ratios 1.0 and 2.0, especially at lower pressures. For mixtures with methane-propane ratios of 9:1 similar tendencies can be observed but with an increased error. Furthermore, the agreement becomes unacceptable (error larger than 20%) for temperatures below 900 K and pressures above 15 atm, and for conditions with lower pressures and very high temperatures. For mixtures with methane-propane ratios of 8:2 essentially the same error pattern can be observed but with slightly increased error. This means a reduced region of accuracy and an increased region with unacceptable predictions. A rectangular region with dimensions of 8–50 atm and 900–1800 K can be located where the mechanism RM1 can predict ignition delays within 20% error for mixtures with 0–20% propane content and equivalence ratios of 0.5–2.0, apart from the 20% propane φ = 2.0 mixtures in the proximity of the condition at 8 atm and 1800 K.

The mechanism RM2 was obtained from mechanism RM1 with reaction removal, which minimized the maximum error in simulated ignition delays for the 22 scenarios. SEM-PCAF led to a mechanism which is even more accurate for these conditions. This enhanced overspecialization of the reduced mechanism makes it possible to achieve small reduced mechanisms with SEM methods at the expense of accuracy at other conditions, which were not defined initially as being important. The mechanism RM2 is not able to predict ignition delays for methane at temperatures below 1000 K anymore, while its high temperature–low pressure accuracy improved significantly. It is more accurate than mechanism RM1 for 10% propane mixtures at φ = 1.0 and 2.0 regarding the whole pressure–temperature range, whereas it works less well for $\varphi = 0.5$ mixtures at high temperatures. It is accurate over the whole pressure and temperature domain above 900 K for 10% propane mixtures with $\varphi = 1.0$. For mixtures with 20% propane the same changes can be observed due to reaction removal: expansion of an accurate region for $\varphi = 1.0$ and 2.0 conditions and its shrinkage for $\varphi = 0.5$ conditions. A rectangular region with dimensions of 8–50 atm and 1000–1800 K can be located where mechanism RM1 can predict ignition delays within 20% error for mixtures with 0–20% propane content and equivalence ratios of 0.5–2.0. For mixtures with significant propane content (10–20%) the lower limit of this temperature range goes down to 900 K. To sum up, the reduced mechanisms can predict ignition delays over a much wider range of conditions than they were developed.

6. Conclusions

The simulation error minimization (SEM) methods and the corresponding program package [43,65] were adapted to be applicable to ignition phenomena. The SEM connectivity method (SEM-CM) was extended by combinatorial complementary sets, which can cope better with the difficult endeavour that requires simulation error to be decreased monotonically as the reduced mechanism is being built up. To ease the increased difficulty of multi-scenario problems, a hierarchical strategy was proposed for these cases, which first prepares reduced mechanisms for single scenarios, and then for small number of similar scenarios ("regime") and finally for all the scenarios. The new version of the program package is available on the World Wide Web [65].

The natural gas mechanism developed at National University of Ireland, Galway (NUIG NGM, 229 species, 1359 reactions) was successfully reduced at the conditions of 22 selected ignition experiments [11] using the extended SEM methods. These experiments investigated lean ($\varphi = 0.5$) and stoichiometric, 90% methane and 10% propane mixtures diluted in 75% Ar and/or N_2 at gas turbine relevant conditions (7–40 atm and 877–1465 K). Reduced mechanism RM1 (251 reactions) was obtained by species removal with SEM-CM, and mechanism RM2 (183 reactions) was obtained by further reaction removal with SEM-PCAF. They can be simulated significantly faster (45× and 62×), while they reproduce ignition delay times very accurately (within 4.9% and 3.1% errors) and also predict pressure rise well despite the fact that pressure rise error was not a controlling parameter.

Robustness analysis was carried out to determine if the reduced mechanisms can reliably reproduce ignition delays calculated by the full mechanism over other conditions. Despite the fact that the SEM algorithms produce highly scenario-specific reduced mechanisms, they have similar or slightly less accuracy over a much wider range of conditions. For details, please refer to Section 5 in this article and the supplementary material.

Robustness analysis of the reduced mechanisms is just as important as the validation of the detailed mechanism, since it usually expands and confines their applicability range at the same time, thus can promote their use over a wider range of conditions while preventing them from being misused. It is questionable to examine their robustness outside the range of conditions where the full mechanism was validated, since their usage there is just as unjustified as that of the full mechanism.

Acknowledgments

Financial support from the EU Marie Curie Transfer of Knowledge Grant (MKTD-CT-2004-517248) and from Science Foundation Ireland under Grant reference number: 07/RFP/CHEF845 are acknowledged. We thank Gilles Bourque for helpful discussions.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2010.12.011.

References

[1] A.H. Lefebvre, Gas Turbine Combustion, Taylor & Francis, Philadelphia, PA, 1998.
[2] L.J. Spadaccini, M.B. Colket, Prog. Energ. Comb. Sci. 20 (1994) 431–460.
[3] R.M. Flores, V.G. McDonell, G.S. Samuelsen, J. Eng. Gas Turb. Pow. 125 (2001) 701–708.
[4] R.M. Flores, M.M. Miyasato, V.G. McDonell, G.S. Samuelsen, J. Eng. Gas Turb. Pow. 123 (2001) 824–831.
[5] J.D. Naber, D.L. Siebers, S.S. Di Julio, C.K. Westbrook, Combust. Flame 99 (1994) 192–200.
[6] T. Lieuwen, V.G. McDonell, E.L. Petersen, D. Santavicca, J. Eng. Gas Turb. Pow. 130 (2008) 011506.
[7] J.M. Simmie, Prog. Energy Combust. Sci. 29 (2003) 599–634.
[8] E.L. Petersen, D.M. Kalitan, S. Simmons, G. Bourque, H.J. Curran, J.M. Simmie, Proc. Combust. Inst. 31 (2007) 447–454.
[9] S.M. Gallagher, H.J. Curran, W.K. Metcalfe, D. Healy, J.M. Simmie, G. Bourque, Combust. Flame 153 (2008) 316–333.
[10] D. Healy, H.J. Curran, J.M. Simmie, D.M. Kalitan, C.M. Zinner, A.B. Barrett, E.L. Petersen, G. Bourque, Combust. Flame 155 (2008) 441–448.
[11] D. Healy, H.J. Curran, S. Dooley, J.M. Simmie, D.M. Kalitan, E.L. Petersen, G. Bourque, Combust. Flame 155 (2008) 451–461.
[12] G. Bourque, D. Healy, H.J. Curran, C. Zinner, D. Kalitan, J. de Vries, C. Aul, E.L. Petersen in: Ignition and flame speed kinetics of two natural gas blends with high levels of heavier hydrocarbons, ASME Turbo Expo 2008: Power for Land, Sea and Air, Berlin, June 9–13, 2008.
[13] D. Healy, M.M. Kopp, N.L. Polley, E.L. Petersen, G. Bourque, H.J. Curran, Energy Fuels 24 (3) (2010) 1617–1627.
[14] D. Healy, D.M. Kalitan, C.J. Aul, E.L. Petersen, G. Bourque, H.J. Curran, Energy Fuels 24 (3) (2010) 1521–1528.
[15] Reaction mechanisms at the Combustion Chemistry Centre, National University of Ireland, Galway. Available from: <http://c3.nuigalway.ie/mechanisms.html> (07/12/2010).
[16] D. Healy, N.S. Donato, C.J. Aul, E.L. Petersen, C.M. Zinner, G. Bourque, H.J. Curran, Combust. Flame 157 (8) (2010) 1526–1539.
[17] D. Healy, N.S. Donato, C.J. Aul, E.L. Petersen, C.M. Zinner, G. Bourque, H.J. Curran, Combust. Flame 157 (8) (2010) 1540–1551.
[18] N.S. Donato, C.J. Aul, E.L. Petersen, C.M. Zinner, H.J. Curran, G. Bourque, J. Eng. Gas Turb. Pow. 132 (2010) 051502.
[19] CFD Online Available from: <http://www.cfd-online.com/Links/soft.html> (07/12/2010).
[20] T. Lu, C.K. Law, Prog. Energ. Comb. Sci. 35 (2009) 192–215.
[21] FLUENT. ANSYS, Inc.: Canonsburg, PA 15317, USA. Available from: <http://www.fluent.com/software/fluent/index.htm> (07/12/2010).
[22] CHEMKIN-CFD. Reaction Design, Inc.: San Diego, CA 92121, USA. Available from: <http://www.reactiondesign.com/products/open/kinetics_module.html> (07/12/2010).
[23] A.S. Tomlin, T. Turányi, M.J. Pilling, in: R.G. Compton, G. Hancock, M.J. Pilling (Eds.), Comprehensive Chemical Kinetics, vol. 35, Elsevier, Amsterdam, 1997, pp. 293–437.
[24] M.S. Okino, M.L. Mavrovouniotis, Chem. Rev. 98 (1998) 391–408.
[25] G. Li, H. Rabitz, J. Tóth, Chem. Eng. Sci. 49 (1994) 343–361.
[26] G. Li, A.S. Tomlin, H. Rabitz, J. Tóth, J. Chem. Phys. 101 (1994) 1172–1187.
[27] A.S. Tomlin, G. Li, H. Rabitz, J. Tóth, J. Chem. Phys. 101 (1994) 1188–1201.
[28] L.E. Whitehouse, A.S. Tomlin, M.J. Pilling, Atmos. Chem. Phys. 4 (2004) 2057–2081.
[29] K.J. Hughes, M. Fairweather, J.F. Griffiths, R.T. Porter, A.S. Tomlin, Proc. Combust. Inst. 32 (2009) 543–551.
[30] T. Turányi, Comput. Chem. 18 (1994) 45–54.
[31] A. Büki, T. Perger, T. Turányi, U. Mass, J. Math. Chem. 31 (2004) 345–362.
[32] B. Yang, S.B. Pope, Combust. Flame 112 (1998) 85–112.
[33] N.J. Brown, G. Li, M.L. Koszykowski, Int. J. Chem. Kinet. 29 (1997) 393–414.
[34] H. Rabitz, F. Aliş, J. Shorter, K. Shim, Comput. Phys. Commun. 117 (1999) 11–20.
[35] T. Turányi, A.S. Tomlin, M.J. Pilling, J. Phys. Chem. 97 (1993) 163–172.
[36] M. Valorani, F. Creta, D.A. Goussis, J. Lee, H. Najm, Combust. Flame 146 (2006) 29–51.
[37] A. Massias, D. Diamantis, E. Mastorakos, D.A. Goussis, Combust. Theor. Model. 3 (1999) 233–257.
[38] A. Massias, D. Diamantis, E. Mastorakos, D.A. Goussis, Combust. Flame 117 (1999) 685–708.
[39] T.F. Lu, Y.G. Ju, C.K. Law, Combust. Flame 126 (2001) 1445–1455.
[40] U. Mass, S.B. Pope, Combust. Flame 88 (1992) 239–264.
[41] CHEMKIN-PRO. Reaction Design, Inc.: San Diego, CA 92121, USA. Available from: <http://www.reactiondesign.com/products/open/chemkin-pro.html> (07/12/2010).
[42] T. Turányi, New J. Chem. 14 (1990) 795–803.
[43] T. Nagy, T. Turányi, Combust. Flame 156 (2009) 417–428.

[44] I.G. Zsély, T. Nagy, J.M. Simmie, H.J. Curran, Reduction of a detailed kinetic model for the ignition of natural gas mixtures at gas turbine conditions, in: Fourth European Combustion Meeting, Vienna, 2009; Vienna, 2009; P810045.
[45] T. Lu, C.K. Law, Proc. Combust. Inst. 30 (2005) 1333–1341.
[46] P. Pepiot-Desjardins, H. Pitsch, Combust. Flame 154 (2008) 67–81.
[47] K.E. Niemeyer, C.-J. Sung, M.P. Raju, Combust. Flame 157 (2010) 1760–1770.
[48] A.S. Tomlin, M.J. Pilling, T. Turányi, J.H. Merkin, J. Brindley, Combust. Flame 91 (1992) 107–130.
[49] A.S. Tomlin, M.J. Pilling, J.H. Merkin, J. Brindley, N. Burgess, A. Gough, Ind. Eng. Chem. Res. 34 (1995) 3749–3760.
[50] L.E. Whitehouse, A.S. Tomlin, M.J. Pilling, Atmos. Chem. Phys. 4 (2004) 2025–2056.
[51] S. Vajda, P. Valkó, T. Turányi, Int. J. Chem. Kinet. 17 (1985) 55–81.
[52] S. Vajda, T. Turányi, J. Phys. Chem. 90 (1986) 1664–1670.
[53] T. Turányi, T. Bérces, S. Vajda, Int. J. Chem. Kinet. 21 (1989) 83–99.
[54] P. Gokulakrishnan, A.D. Lawrence, P.J. McLellan, E.W. Grandmaison, Comput. Chem. Eng. 30 (2006) 1093–1101.
[55] H. Wang, M. Frenklach, Combust. Flame 87 (1991) 365–370.
[56] B. Bhattacharjee, D.A. Schwer, P.I. Barton, W.H. Green, Combust. Flame 135 (2003) 191–208.
[57] L. Elliott, D.B. Ingham, A.G. Kyne, N.S. Mera, M. Pourkashanian, C.W. Wilson, Ind. Eng. Chem. Res. 44 (2005) 658–667.
[58] I. Banerjee, M.G. Ierapetritou, Combust. Flame 144 (2006) 619–633.
[59] O.O. Oluwole, P.I. Barton, W.H. Green, Combust. Theor. Model. 11 (2007) 127–146.
[60] L. Liang, J.G. Stevens, S. Raman, J.T. Farrell, Combust. Flame 156 (2009) 1493–1502.
[61] K. He, I.P. Androulakis, M.G. Ierapetritou, Chem. Eng. Sci. 65 (2010) 1173–1184.
[62] T. Lu, C.K. Law, Combust. Flame 144 (2006) 24–36.
[63] X.L. Zheng, T. Lu, C.K. Law, Proc. Combust. Inst. 31 (2007) 367–375.
[64] J. Revel, J.C. Boettner, M. Cathonnet, J.S. Bachman, J. Chim. Phys. 91 (1994) 365–382.
[65] T. Nagy, T. Turányi SEM: mechanism reduction based on simulation error minimization. Available from: <http://garfield.chem.elte.hu/Combustion/sem.htm> (07/12/2010).
[66] I. Banerjee, M.G. Ierapetritou, Ind. Eng. Chem. Res. 44 (2005) 3638–3647.
[67] O.O. Oluwole, B. Bhattacharjee, J.E. Tolsma, P.I. Barton, W.H. Green, Combust. Flame 146 (2006) 348–365.
[68] T. Turányi, I.G. Zsély, KINALC: a CHEMKIN based program for kinetic analysis. Available from: <http://garfield.chem.elte.hu/Combustion/kinalc.html> (07/12/2010).
[69] T. Turányi, Mechmod v.1.4: Program for the transformation of kinetic mechanisms. Available from: <http://garfield.chem.elte.hu/Combustion/mechmod.htm> (07/12/2010).